Read Trimming and Adapter Removal

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Introduction

After FASTQ-level QC has identified low-quality bases and adapter contamination, read trimming removes them before alignment. The motivation is straightforward: sequencing-quality scores typically degrade toward the 3' end of reads, and adapter sequences appear when the insert is shorter than the read length. Both produce mismatched bases that bias alignment and increase mapping rates incorrectly. The `Rfastp` package provides an R-friendly interface to `fastp`, the fast C++ tool that handles trimming, adapter removal, quality filtering, and polytail cleanup in a single pass.

Prerequisites

A working understanding of FASTQ format, Phred quality scores, and Illumina adapter sequences (TruSeq, Nextera).

Theory

Standard trimming operations:

- **Quality trimming**: scan from the 3' end (and optionally 5') and remove bases falling below a Phred threshold; sliding-window variants trim when the mean quality in a window drops below a cutoff.
- **Adapter removal**: detect known adapter sequences via overlap matching with the read 3' end and trim them out. Auto-detection (matching to a library of common adapters) is usually reliable.
- **Length filtering**: discard reads that fall below a minimum length after trimming; very short reads align ambiguously and waste downstream resources.
- **Polytail trimming**: remove poly-A or poly-G contamination, common in NextSeq/NovaSeq dark-cycle artefacts.

Assumptions

Adapter sequences are known or auto-detectable; quality scores are on the standard Phred+33 encoding (true for all modern Illumina output).

R Implementation

```r
library(Rfastp)

in_file  <- system.file("extdata", "reads1.fastq.gz", package = "Rfastp")
out_file <- tempfile(fileext = ".fastq.gz")

# Trim with fastp default heuristics
out <- rfastp(read1 = in_file, outputFastq = out_file,
              thread = 2,
              trimAdapter = TRUE, overlapDiff = 5)

out$summary
```

Output & Results

`rfastp()` returns a summary list with before/after statistics: reads passing filter, reads with adapters trimmed, mean read length before and after, base content. The HTML report (auto-generated by `fastp`) provides a richer visualisation of the trimming impact.

Interpretation

A reporting sentence: "Read trimming with fastp removed Illumina TruSeq adapters from 3.2 % of reads and quality-trimmed bases below Phred 15; 98 % of reads were retained after trimming, with mean read length decreasing from 150 to 145 bp." Always report the adapter set, quality threshold, and retention rate.

Practical Tips

- Keep trimming settings mild — aggressive quality filtering removes data without proportional gains in downstream accuracy.
- Adapter auto-detection in `fastp` is usually reliable; specify adapter sequences explicitly only when working with non-standard library prep kits.
- For paired-end reads, trim both mates simultaneously and retain only properly-paired survivors; `Rfastp::rfastp()` with `read1` and `read2` arguments handles this automatically.
- Validate downstream QC (FastQC, MultiQC) after trimming; a successful trim improves per-base quality, per-base content, and adapter-content panels.
- Record trimming parameters in a methods document or workflow file (Snakemake, Nextflow); reproducibility requires exact tool versions and parameters.
- For UMI-tagged libraries (single-cell, low-input), use UMI-aware tools (`umi_tools`, fastp's UMI mode) before quality trimming to preserve UMI integrity.

R Packages Used

`Rfastp` for the canonical fastp wrapper; `ShortRead` for general FASTQ manipulation in R; `Biostrings` for adapter-sequence handling; `QuasR` for an integrated alignment workflow that includes trimming as a step.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat